

# AKSHAY GOVIND

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## EDUCATION

**Northeastern University**, Boston, MA

Sep 2025 – Dec 2027

*Master of Science, Information Systems*

Relevant Coursework: Data Management and Database Design, Designing Advanced Data Architecture for Business Intelligence, Data Science Engineering Methods and Tools, Theory and Practical Applications of AI Generative Modeling

**University of Mumbai**, Mumbai, India

May 2022

*Bachelor of Engineering, Computer Engineering*

## SKILLS

**Languages & Frameworks:** SQL, PostgreSQL, Oracle, Python, Pandas, NumPy, Scikit-learn, PySpark

**Machine Learning & AI:** Machine Learning, Generative AI, Large Language Models (LLMs)

**Data Engineering & Ops:** Databricks, Azure Data Factory, Delta Lake, Git, GitHub, Jira

**Cloud & Warehousing:** Snowflake

**Analytics & Reporting:** Tableau, Power BI, Streamlit, Data Visualization, Statistical Analysis

## EXPERIENCE

**Vaishnav Publications and Digitalization**, Pune, India

Jan 2023 – Apr 2025

*Business Intelligence Analyst*

- Designed and maintained 8+ Power BI dashboards tracking sales and operational KPIs, enabling business stakeholders to monitor performance trends and make data-driven decisions during weekly reviews.
- Diagnosed and resolved data quality issues, including inconsistent formatting, mismatched fields, and currency discrepancies, across a 10,000+ row multi-client dataset to improve reporting reliability.
- Automated a manual weekly reporting workflow using Python and SQL, reducing report preparation time from up to 4 hours to under 30 minutes while improving reporting accuracy.
- Gathered and analyzed business requirements from cross-functional stakeholders, translated business needs into Jira user stories and acceptance criteria, and collaborated with Agile teams throughout sprint planning and delivery.

## PROJECTS

### Instacart Analytics Platform

*PostgreSQL (Neon), Python, Streamlit, Plotly, SQL, Advanced Analytics (RFM)* | [GitHub](#) | [Live Demo](#)

- Architected a relational database schema using cloud-hosted PostgreSQL to ingest a 33M+ row benchmark dataset, modeling transactional grocery data across core relational tables.
- Developed analytical scripts spanning window functions, stored procedures, and triggers to perform customer RFM segmentation, cohort retention analysis, and CLV tiering.
- Optimized query performance by analyzing execution plans via EXPLAIN ANALYZE and deploying composite and partial indexes, then built a live Streamlit dashboard for interactive exploration.

### Wildfire Cause Prediction

*Python, Scikit-learn, Pandas, NumPy, Machine Learning (Random Forest, SVM, KNN), Feature Engineering* | [GitHub](#)

- Developed a binary classification framework on a 1.88M-row USDA Forest Service fire occurrence database to predict human-caused vs. natural/lightning-caused wildfires.
- Engineered 5 spatial and temporal features, including historical state fire rates and high-risk drought indexing, to overcome a severe 5.2:1 class imbalance.
- Evaluated and cross-validated 5 machine learning models using statistical analysis techniques, training on a stratified 80/20 split and tuning Random Forest via GridSearchCV to a peak F1 = 0.9598 and ROC-AUC = 0.9547.

### FoodLens – Food Inspections ELT Pipeline

*Databricks, PySpark, Delta Lake, Unity Catalog, SQL, Power BI, Kimball Star Schema* | [GitHub](#)

- Engineered a Medallion architecture pipeline across Bronze, Silver, and Gold layers using Databricks and Unity Catalog to ingest and process 387,000+ structurally inconsistent municipal food inspection records.
- Resolved severe schema mismatches by writing PySpark transformation logic to unify divergent formats (pipe-delimited strings vs. 25 distinct columns) using MD5 hashing for consistent cross-source surrogate keys.
- Constructed an optimized star schema using Delta Lake MERGE operations for SCD Type 2 dimension tracking, while deploying a custom validation engine that automatically quarantined corrupted rows.